

separation of reusable and recyclable components, leading to ramshackle piles of dismembered pieces of discarded electronic equipment

Mechanical recycling.

Removal of hazardous components followed by granulating and shredding, in order to remove recyclable raw materials such as plastics and ferrous metals

Chemical stripping. Precious metals such as gold and silver are recovered from gold-plated chips, components and printed circuit boards using nitric and hydrochloric acids

Open burning. Metals such as copper, nickel, aluminium and tin alloys are recovered from the most combustible materials by open burning method using wood, charcoal, etc

These rudimentary methods are found to cause soil acidification, emission of toxic fumes and gases, leaching into ground water and soil and thus its poisoning, etc.

Disposing e-waste

The present disposal practices are often dangerous and unsatisfactory. Thousands of abandoned, often unknown and unorganised, waste disposal sites are leaving toxic materials in nature, posing significant health risks. Alternative techniques for disposing the e-waste include:

Physical treatment. Physical treatments tie up or isolate substances. Charcoal or resin filters absorb toxins. By precipitation and immobilisation, ceramics and glass isolate toxins from the environment so the hazardous materials become essentially non-hazardous. One of the few ways to dispose of metals and radioactive substances is to fuse them in silica at high temperatures to make a stable, impermeable glass that is suitable for eco-friendly, long-term disposal and storage.

Chemical processing. Chemical processing can transform materials so



e-waste recycling

that these become non-toxic. Included in this category are the processes based on neutralisation, electrolysis, filtration and also removal of metals and halogens based on reverse osmosis and hydrometallurgy.

Landfill. Inevitably, there are some materials that cannot be destroyed. These have to be stored out of harm's way. One such way is secure landfills. A landfill, also known as a dump or tip, is a site for disposal of waste materials by burial—the oldest form of waste treatment. Though this method of waste disposal is simple, care must be taken so that toxic substances don't leach into the soil. The storage area must be guarded and monitored continuously to prevent leakage, vandalism or other dispersal of toxic materials.

Incineration. Incineration is a waste treatment technology that involves combustion of organic materials/substances. It is applicable to mixtures of wastes. Improper combustion of e-waste due to low temperature can lead to toxic emissions containing polychlorinated biphenyls, polybrominated biphenyls, polybrominated biphenyl ethers, etc.

Owing to rapid improvement in the incineration technology, today's incinerators are backed with pollution mitigation equipment that filter and

clear the flue gases for pollutants before they are dispersed in the atmosphere. Liquid injection nozzles and fluidised bed burners ensure quick and complete combustion of e-waste during incineration process.

Dealing with e-waste

It's high time for sustainable production with justice to the environment and corporate and government accountability

in order to achieve an effective way to deal with e-waste. Strategies must be evolved to reduce generation of e-waste, prevent legal or illegal imports of such waste, and develop feasible and safe ways of dealing with it within the present context and requirements. This can be achieved with involvement and close association of major stakeholders (government, public-private sectors and industries) and society at large.

Responsibility of government. The government must play a vital role as a guiding force in dealing with e-waste. It should provide an adequate system of laws, controls and administrative procedures for hazardous waste management. It must come forward to:

1. Frame a comprehensive law that provides an effective e-waste regulation and management system
2. Encourage advancement of research to invent cutting-edge technologies for handling e-waste
3. Devise ways and means to motivate beneficial reuse/recycling of e-waste, catalysing business activities related to the same
4. Enforce strict regulations and heavy fines on companies that do not practice waste prevention and recovery in their production facilities
5. Review the trade policy and EXIM classification codes to plug the